

# ELI — ELI USER MANUAL

ELI v2.1.35

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## ELI — The Complete User Manual

### Plain-English Edition · with flowcharts

*A friendly, no-jargon guide to everything ELI can do and how to make it do it.*

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This manual assumes **zero** technical background. You talk to ELI in normal English — by **typing** in the chat box or **speaking** — and it does things on your computer, answers questions, remembers what matters to you, and learns your routines. Everything runs **on your own machine**; nothing is sent to the cloud.

**The one rule worth knowing:** ELI is **offline by default**. It only touches the internet when you ask for something that needs it (a web search, the news, downloading a model). Your

conversations, files, and memories never leave your computer.

How to read this manual: - **Just want to use it?** Read §1–§5 and the cheat sheet (§18). That’s enough to be productive. - **Want the whole picture?** Read straight through — every feature is here, in plain terms. - **[Everyone]** / **[Some setup]** / **[Advanced]** tags at the start of some sections mean *anyone* / *a little technical* / *advanced*.

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## 1. What ELI is (in one minute)

ELI is a **personal AI assistant that lives on your computer**. Think of it as a capable helper that can:

- **Talk with you** like a knowledgeable person — by text or voice.
- **Operate your computer** — open apps, play music, manage windows, type for you.
- **Look at things** — your screen, images, PDFs, spreadsheets — and explain them.
- **Write for you** — documents, notes, code, scripts.
- **Remember** what matters about you, and **learn** your daily routines.
- **Act on its own** when helpful — a morning briefing, a heads-up, a suggestion.

The big difference from cloud assistants: **it’s yours**. It runs locally, stays private, has no subscription, and you can even retrain its “voice” yourself (§14).

Here’s the whole thing at a glance:

```
mindmap
  root((ELI))
    Talk
      Type or speak
      Wake word
    Do
```

- Open apps
- Play music
- Manage windows
- Type for you
- See
  - Your screen
  - Images and PDFs
  - Spreadsheets
- Create
  - Documents
  - Notes
  - Code and scripts
- Know you
  - Memory of facts
  - Living profile
  - Learned habits
- Act on its own
  - Morning briefing
  - Suggestions
  - Overnight tasks
- Private
  - Runs locally
  - Offline by default
  - No accounts

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## 2. How ELI works when you ask it something

[Everyone] *Anyone — this is the mental model that makes everything else click.*

When you say or type something, ELI follows the same honest loop every time:

flowchart TD

```
A([You ask ELI something<br/>type or speak]) --> B{Is it a command<br/>or a conversation?}
B -->|A command<br/>'open spotify'| C{Does it need<br/>the internet?}
B -->|A conversation<br/>'how are you?'| H[ELI replies in its<br/>own voice]
C -->|No| D[ELI does it<br/>on your computer]
C -->|Yes, e.g. news/search| E{You allowed<br/>online for this?}
E -->|Yes| F[ELI fetches it,<br/>tells you it went online]
E -->|No| G[ELI stays offline<br/>and says so]
D --> I{Did it actually<br/>work?}
F --> I
I -->|Yes| J([ELI confirms what it did])
I -->|No / unsure| K([ELI tells you honestly —<br/>never pretends it worked])
H --> J
```

**The promise in that diagram:** ELI **never fakes an action**. If it says it played a song or fixed a file, it really did — and if it couldn't, it says so plainly instead of bluffing. This is built into the software, not just good manners.

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## 3. Getting started

[Some setup] *A little technical, but only once.*

## Installing (v2.1.23 and later — one download per platform)

Get the latest from **GitHub Releases**. Every release is built in CI and **launch-tested on Windows, macOS and Linux** before it can publish.

- **Windows:** run `ELI-Setup-<version>.exe` — installs per-user, no admin password. If it sees an NVIDIA card it offers GPU acceleration during setup; if it finds an older ELI it offers a **fresh install** (resets settings & memory; keeps downloaded models). You get three shortcuts: **ELI**, **ELI Server**, **Uninstall ELI**.
- **macOS (Apple Silicon):** open the .dmg, drag **ELI** into Applications. First launch is right-click → **Open** (the app is unsigned). Apple-GPU acceleration is built in — nothing to set up.
- **Linux:** `chmod +x ELI_v2-<version>-x86_64.AppImage` and run it. On first launch it offers **applications-menu entries** (ELI, ELI Server, and a working Uninstall).

flowchart TD

```
A([Run the installer / AppImage]) --> B[First launch]
B --> C{GPU detected?}
C -->|NVIDIA| D[Offer: CUDA engine<br/>one-time download, verified]
C -->|AMD / Intel Arc| E[Offer: Vulkan engine<br/>one-time download, verified]
C -->|Apple| F[Metal built in]
C -->|None / decline| G[CPU engine - always works]
D --> H{Chat model installed?}
E --> H
F --> H
G --> H
H -->|No| I[Offer: starter model<br/>sized to your hardware]
H -->|Yes| J
I --> J[Quick interview]
J --> K([Ready - say hello])
```

## What arrives on first launch

| Asset                                    | When                          | Size                                      |
|------------------------------------------|-------------------------------|-------------------------------------------|
| <b>Piper voices</b> (speech out)         | already inside the installer  | —                                         |
| <b>Nomic embedder</b><br>(memory/recall) | fetches automatically         | ~80 MiB                                   |
| <b>Whisper</b> (speech in)               | fetches on first voice use    | ~460 MiB                                  |
| <b>GPU engine</b> (optional)             | if you accept the GPU offer   | ~90 MiB (AMD) · ~1-2 GiB (NVIDIA)         |
| <b>Chat model</b>                        | if you accept the model offer | sized to your hardware<br>(e.g. ~4.7 GiB) |

Everything ELI knows and downloads lives in **one per-user folder** that **survives upgrades** — like a save-game: `%LOCALAPPDATA%\ELI_v2` (Windows), `~/Library/Application Support/ELI_v2` (macOS), `~/.local/share/ELI_v2` (Linux). Databases start as a **true blank slate** — no personal data ships in any installer.

## Starting over, or uninstalling

- **Clean slate:** tick *Fresh install* in the Windows setup, or run `ELI --fresh-start` anywhere (keeps downloaded models unless you add `--purge-models`).
- **Uninstall:** Windows → the **Uninstall ELI** shortcut. Linux → the **Uninstall ELI** menu entry (removes the entries, offers to delete your data, then delete the AppImage file). macOS → drag the app to Trash; your data folder is listed above.
- **GPU later / undo:** `ELI --install-gpu-pack` (add `--vulkan` for Intel Arc) · `ELI --remove-gpu-pack`.

- **From source (developers):** `bash install.sh` from a clone works exactly as before.

### First launch — the quick interview

The very first time you open ELI, it doesn't know you yet, so it asks **three short questions** (your name, what you do, how you like to be spoken to). Ten seconds, and you can **skip it** entirely. From there, ELI builds its understanding of you naturally as you talk.

### Opening ELI

- **Desktop app (full experience):** the **ELI** shortcut / menu entry (installer builds), the `ApplImage` itself (Linux), or `scripts/eli_launch.sh` from a source clone.
  - **From your phone:** the **ELI Server** shortcut (or `ELI --server`) starts the home server with the connect QR — see §13.
- 

## 4. Your first hour with ELI

[Everyone] A gentle on-ramp. Try these in order — each builds confidence.

1. **Say hello.** Type `how are you?` — get a feel for ELI's voice.
2. **Ask what it can do.** `what can you do?` lists its capabilities.
3. **Open something.** `open the file manager` or `open firefox`.
4. **Play music.** `play lo-fi` on YouTube, then pause, then skip.
5. **Look at your screen.** `what's on my screen?`
6. **Tell it a fact.** `remember that my dog's name is Rufus`. Later: `what's my dog's name?`
7. **Set a reminder.** `set a timer for 5 minutes`.
8. **Get a briefing.** `what's the news?` (this one goes online — ELI will say so).
9. **Check on ELI itself.** `what are you running on?` shows its model and hardware.
10. **Go hands-free.** Say the wake word **“computer”**, then a request.

By the end you've touched conversation, control, vision, memory, reminders, the web, voice, and introspection — the whole surface.

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## 5. Talking to ELI

There are two ways, freely mixed:

- **Type** in the chat box and press Enter.
- **Speak** — click the microphone, or say the **wake word** (default **“computer”**) then your request. Change the wake word to anything (§11).

You don't need special phrasing. Talk normally:

`“open spotify and play some lo-fi” “what's on my screen?” “remember that my sister's name is Anna” “set an alarm for 7am”`

**Chaining:** ask for several things at once — `> “close steam and set an alarm for 7am” > “open spotify then play Juicy by Notorious B.I.G.”`

**If ELI isn't sure** what you meant, it asks — it won't guess wildly or pretend. And if a request needs the internet, it tells you before going online.

**If you're having a hard time:** ELI is tuned to notice genuine first-person distress (it's built to ignore ambient audio and dark jokes, not to over-react). When it does, it steps out of task-mode and gently checks in — pointing you toward real human support rather than ploughing on with commands or reciting a canned line. It's a steer, not a script.

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## 6. A tour of the window

[Everyone] ELI Pro is organised into **tabs**. You'll mostly live in **Chat**; here's the rest in plain terms:

| Tab                   | What it's for                                                                                                                              |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Chat</b>           | Talk to ELI, by voice or text. Home base.                                                                                                  |
| <b>Quick Actions</b>  | One-tap buttons for the things you do most.                                                                                                |
| <b>Memory</b>         | Browse what ELI remembers about you; add or remove facts.                                                                                  |
| <b>Habits</b>         | Routines ELI has learned; turn on/off, edit times, add your own.                                                                           |
| <b>Proactive</b>      | Controls for ELI's "act on its own" behaviour (suggestions, summaries, insights).                                                          |
| <b>Tasks</b>          | Scheduled & overnight jobs and their status.                                                                                               |
| <b>Images</b>         | Generate images — procedural (fast) or diffusion (photoreal).                                                                              |
| <b>Screen</b>         | Watch/read your screen, take screenshots, OCR.                                                                                             |
| <b>Coding / IDE</b>   | Write, examine, and fix code; a built-in editor.                                                                                           |
| <b>Self-Improve</b>   | What ELI is proposing to improve in <i>itself</i> — approval-gated.                                                                        |
| <b>Report Builder</b> | Generate longer documents/reports with sources (evidence → outline → draft → revise).                                                      |
| <b>Elis World</b>     | ELI's own world-model view.                                                                                                                |
| <b>Labs</b>           | Power-user workshop — notebook, Jupyter, calculator, physics, file-chat, <b>projects/workspaces</b> , sim-IDE, orchestration, test review. |
| <b>Settings</b>       | Model (incl. the model-switch dropdown), hardware, voice, and privacy options (\$16).                                                      |

You rarely need the tabs directly — almost everything is reachable by **asking** in Chat: "show my habits", "what do you remember about me", "show background jobs", "make an image of ...", and ELI opens or reports the right thing.

---

## 7. What ELI can do — by everyday task

[Everyone] Real things you can say. ELI understands many phrasings — these are examples.

### Music & media

- "play lo-fi on YouTube" · "play Juicy by Notorious B.I.G. on Spotify"
- "pause" · "skip" · "next song" · "previous track" · "shuffle" · "repeat this song"
- "what's playing?" · "stop the music" · "skip the ad" (YouTube, when skippable)

### Apps & windows

- "open spotify" · "launch firefox" · "open the file manager" · "open github.com"
- "close chrome" · "focus the browser" · "switch to workspace 2"
- "tile windows" · "maximise the window" · "show desktop" · "next window"

## Volume, typing, mouse

- “volume up” · “mute” · “set volume to 30”
- “type hello world” · “press enter” · “left click” · “move the mouse up”

## Files & notes

- “create a file notes.txt” · “make a folder called projects” · “read notes.txt”
- “list the files in ~/Documents” · “write a note saying buy milk” · “search notes for budget”
- “what’s in my clipboard?” · “summarise report.docx” · “convert report.md to pdf”

## Writing, documents & code

- “write a report on renewable energy” · “generate a document about X”
- “write a bash script to monitor the GPU” · “solve this: implement a function that ...”
- “fix the bugs in foo.py” · “examine eli/memory/memory.py for errors” → ELI scans, **offers** fixes, and applies them only if you say “yes, fix it”. · “show the diff”
- “fabricate a test dataset for X” → ELI generates synthetic/test data on a topic.
- The **Report Builder** tab turns “generate a document on X” into a real report: it gathers evidence, plans an outline, drafts each section, then **reviews and revises** it — with document-type quality profiles. It’s generation-first, not a one-shot dump.

## Screen, images, PDFs, spreadsheets

- “what’s on my screen?” · “find the submit button on screen” · “take a screenshot”
- “describe cat.jpg” · “read the text in image.jpg” (OCR)
- “summarise report.pdf” · “analyse data.csv” · “watch my screen” / “stop watching”

## Information

- “what time is it?” · “what’s the date?” · “weather in Wexford”
- “what’s the news?” / “catch me up” → a **synthesised** briefing, not a raw dump
- “search the web for X” (goes online — ELI tells you) · “gpu status” · “system stats”

## ☐ Reminders, timers, calendar, scheduling

- “set an alarm for 7am” · “set a timer for 10 minutes”
- “add an event tomorrow at 3pm” · “list my events” · “start a pomodoro”
- “research the best solar inverters overnight” · “build me a script at 2am” → **background jobs**
- “show background jobs” · “check job 5”

## Asking ELI about itself (honest introspection)

- “what are you running on?” (model, context, GPU) · “how does your memory work?”
- “what do you know about me?” · “what have you been working on?” · “status”

## Eye control (gaze)

- “calibrate gaze”, then “enable gaze”. With it on, ELI clicks **where you’re looking** when you say “open” / “left click” / “right click” / “hit enter” — hands-free pointing.
- **Fully hands-free and voice-free — dwell-click.** Say “**enable gaze clicking**” (or turn on **Dwell-click** in Settings) and ELI clicks wherever you **rest your gaze for about a second** — no mouse, no voice needed. This is built for people who can’t use their hands *or* speak reliably. Tune the hold time, wander radius, and sensitivity in Settings (gaze\_dwell\_seconds, gaze\_dwell\_radius). Turn it off any time; it’s opt-in.
- “gaze status” · “disable gaze”

## Making images

- “make an image of a red bike by the sea” · “generate a picture of ...”
- Two engines: a fast **procedural** one (always available) and **diffusion** (SSD-1B) for photoreal results — ELI briefly frees the model’s VRAM to run it. The **Images** tab has the controls; results save locally.

## Coding & building things

- “solve this: implement a function that ...” → ELI plans it, writes it, then **verifies and repairs** it.
- “generate a project that does X” (a full multi-file scaffold) · “write a bash script to ...”
- “examine eli/memory/memory.py for errors” → a tiered scan; it **offers** fixes and applies them only if you say “yes”. · “show the diff” · “generate tests for your code” (it writes *and* sandbox-verifies them). The **Coding** tab and the built-in **IDE** are home for this.

## ELI improving itself

- “improve yourself” → a self-improvement patch cycle — proposed, and **gated by your approval**.
- “show your self-improvement log” · “patch yourself” · “upgrade yourself” (git pull → deps → rebuild indexes) · “run your self-tests”. The **Self-Improve** tab shows what it’s proposing.

## Plugins & your own agents

- “list plugins” · “enable the X plugin” · “install the X plugin” · “plugin status”
- You can **build your own agent** just by describing it — ELI runs a short dialog, validates it, and registers it live (the create-agent flow).

## Overnight & scheduled work

- “research the best solar inverters overnight” · “build me a script at 2am” · “every night, regenerate the test report” → durable **background jobs** that survive restarts.
- “show background jobs” · “check job 5”. The **Tasks** tab lists them.

## Shaping ELI’s voice (persona)

ELI’s personality is **emergent** — it forms from your profile, your history, and the way you talk to it, not from a hand-written script. Most of the time you leave it alone and it just *becomes* itself with you. When you want to steer it, you can, without erasing that: - “lock your persona to a terse senior engineer” · “what persona are you locked to?” · “clear persona lock”. A lock **layers on top** of the emergent voice for the session; clearing it hands the wheel back to the natural persona. - “refresh your persona” reloads it from your latest profile (after ELI’s learned something new about you). - **Set how it talks to you** (*new, 2.1.23*) — say “*speak to me more bluntly*”, “*be funnier*”, or “*respond as Rick*” and it **sticks** (say “*reset your tone*” to clear it). You can also set it in **Settings → Identity → Tone / how ELI speaks**. This explicit preference takes priority over the tone ELI infers, but leaving it blank keeps ELI’s natural emergent voice in charge. - ELI also keeps a small set of **core values** (memory policy, verbosity, safety) that are part of its persona and evolve alongside it.

The rule of thumb: **the emergent voice is the default and the truth of who ELI is with you; a lock is a temporary hat it wears on request**. ELI won’t drift into a scripted character on its own, and a lock never overwrites the underlying personality — it sits on top until you clear it.

## Tone & emotion — reading the room

ELI now **reads the emotional situation from two independent signals** — the *sound* of your voice (energy, mood) and the *content* of what you say — and shades **how** it delivers: its wording, its voice (pace, warmth, pauses), and its avatar expression all move together. If you’re clearly frustrated it gets calmer and more direct; if you’re playing, it plays back; if you’re upset, it softens. **This never changes**



**who ELI is** — its core personality stays exactly the same; it just picks the tone that’s most useful for the moment.

You can also set the register yourself: “*be comedic*”, “*talk street*”, “*sound more professional*”, “*use a deadpan tone*” — and “*go back to normal*” (or “*be yourself*”) to hand it back to ELI’s own read of the room. There’s a growing palette of tones (comedic, professional, curious, deadpan, tender, street-smart, gremlin, and more), and you can add your own in config/voices/tones.json. Turn the whole thing off with ELI\_TONE\_ADAPT=0.

---

## 8. ELI remembers you (memory)

[Everyone] ELI builds a private, evolving understanding of **who you are** and reads it every time you talk, so you never repeat yourself.

flowchart TD

```
A([You're talking to ELI]) --> B{What kind of thing<br/>did you say?}
B -->|Remember that ...'| C[Saved as a durable fact<br/>kept until you remove it]
B -->|Just chatting<br/>interests, focus, tone| D[Quietly updates ELI's<br/>living profile of you]
C --> E[(Local memory<br/>on your computer)]
D --> E
E --> F([Read every turn so ELI<br/>stays in context])
F --> G[/Old, abandoned topics<br/>fade over time/] --> E
```

**Tell it something to keep:** “remember that my sister’s name is Anna” · “my name is Alex” **Ask what it knows:** “what do you remember about my car?” · “what do you know about me?” · “explain everything you know about me and where it’s stored” (a sourced, detailed answer).

**It remembers the thread, not just facts:** when a conversation ends, ELI quietly summarises it and carries a digest of your recent sessions forward — so it recalls what you’ve been *working through* over days, not just isolated one-off facts.

Everything is stored **locally** in small private databases. Review and prune it in the **Memory** tab. A brand-new install is a **blank slate** — ELI knows nothing about you until you talk.

---

## 9. ELI learns your routines (habits)

[Everyone] If you do the same thing at roughly the same time on **several different days**, ELI notices and **offers** to automate it. It never activates anything without your “yes”.

flowchart TD

```
A[You open your email app<br/>around 9am, several days] --> B{Same action, same time,<br/>on 3+ different days}
B -->|No| X[ELI stays quiet<br/>no nonsense suggestions]
B -->|Yes| C[ELI offers:<br/>'Add this as a habit? yes/no']
C --> D{Your answer}
D -->|No| E[Dropped, won't ask again]
D -->|Yes| F{Has today's time<br/>already passed?}
F -->|Yes| G[Runs it now once,<br/>then daily from tomorrow]
F -->|No| H[Runs at the time, daily]
G --> I([Active habit – edit in Habits tab])
H --> I
```

Manage everything in the **Habits** tab (add, edit times, turn off), or say “show my habits”. The key safeguards: it only proposes routines you’ve **genuinely repeated across days**, and it **always asks first**.

---

## 10. ELI helps before you ask (proactive)

[Everyone] With **proactive mode** on, ELI can do helpful things on its own — a **morning briefing**, a gentle suggestion, or a goal it thinks would help. It's conservative and easy to control:

- “start proactive mode” / “stop proactive mode” · “proactive status”
- “morning report” / “give me my briefing” (news, weather, your agenda)
- “do you have any proposals for me?” / “what’s on your agenda?”

Schedule real unattended work too: > “research X overnight” · “generate tests for your code tonight”

These appear in the **Tasks** tab and survive restarts.

### The autonomy tick — what “on its own” actually means

When you allow it, ELI runs a quiet **autonomy tick** inside the proactive daemon — roughly every half hour, and only if it's switched on (kill switch: `ELI_AUTONOMY_TICK=0`). One tick does three things, all **read-only** or **proposal-only**:

1. **Code monitor** — it glances over its own code health and flags anything that's drifted.
2. **Self-model refresh** — it updates its understanding of itself (what it's running, what it can do) so its introspection stays honest.
3. **Goal autogenesis** — it may propose a goal, or a scheduled task, it reckons would help you.

The guardrail that matters: every autogenerated goal is **proposal-only**. It shows up as a suggestion you accept or dismiss — nothing destructive ever runs unattended, and anything risky still goes through the **approval gate** (Appendix C). Autonomy makes ELI *thoughtful* on its own, not *loose*.

---

## 11. Voice, wake word, and dictation

[Everyone] ELI is fully usable hands-free.

flowchart LR

```
A([Say the wake word<br/>'computer']) --> B[ELI starts listening]
B --> C[You speak your request]
C --> D[ELI transcribes it<br/>on your computer]
D --> E[ELI does the task]
E --> F([Optionally speaks the answer back])
```

- **Wake word:** say “**computer**” then your request. Change it: “change the wake word to athena”. ELI trains its listener on the new word automatically — works even over background music.
- **Personalise to your voice:** “enroll my wake word” (records you saying it and adapts).
- **Dictation:** “start dictation” / “stop dictating” — speak and ELI types into the active field.
- **Transcribe a recording:** “transcribe audio.wav”
- **ELI speaking back:** “say good morning”
- **A human-like voice:** at the **startup model picker** (before ELI loads its model) pick **Voice engine** → “**Natural neural (XTTS-v2)**” for an expressive, human-sounding voice — or say “*use the natural voice*” any time. It needs a one-time `pip install "eli-v2.0[natural]"` (downloads ~1.8 GB on first use) and a reasonable GPU; on a lighter machine leave it on **Piper**, which is fast and now noticeably more natural (it pauses on full stops and lifts on questions).
- **Change ELI's voice:** the voice dropdown next to the chat box, **Settings > Voice / TTS**, or just ask — “*use the alan voice*”, “*switch your voice to amy*”, “*use a natural male voice*”.
- **Get more voices and accents:** ask ELI directly — “*what voices do you have*”, “*download a British voice*”, “*get me a Scottish accent*” — or use **Settings > Voice / TTS > “Get more voices / accents...”**. Either way, ELI can download from a library of **166 voices across 45 languages** — 38 of them English (US, British, Scottish, Northern and Southern English). Each is verified, installed, and selectable immediately. A few voices are marked *personal use only*: their licences don't allow ELI to bundle them, but you can still download them yourself.

- **Character voices:** choose `char:hal`, `char:tars`, `char:rick`, `char:glados` or `char:jarvis` from the same dropdown — each is a real voice plus a live effect chain. You can make your own, and down-loading a character’s “ideal base” voice (flagged in the voice library) sharpens it.
- **Make your own voice:** **Settings > Voice / TTS** — either **drop in a clip** (a short, clean ~6-20s .wav/.mp3/.mp4 of the voice you want) or click **“Record a voice now”** to capture 12 seconds from your mic on the spot. You can also say *“create a voice called Nova from clip.mp4”*. It sounds like your sample once the neural voice extra is installed (`pip install "eli-v2.0[natural]"`); until then it’s saved and ELI uses a normal voice.
- **ELI’s face reacts:** the little animated face shows how ELI feels, **moves its mouth while it speaks**, and shows a focused look while it’s thinking.
- **Teach ELI your voice & tone:** “train my voice” — it learns your pitch/energy and even emotion cues, then adapts how it speaks to you.
- **If voice misbehaves:** “run voice diagnostics”.

---

## 12. Seeing your screen and images (vision)

[Everyone] ELI can actually **look**:

- **Your screen:** “what’s on my screen?” · “what does this error say?” · “find the submit button”
- **Image files:** “describe cat.jpg” · “read the text in this image” (OCR)
- **Documents:** “summarise report.pdf” · “analyse data.csv” · “analyse the pdfs in that folder”
- **Continuous watching:** “watch my screen” (and “stop watching”)

The first time you use vision, ELI may fetch a small vision model (you can pre-download it — \$14). With a webcam set up, ELI even supports **eye-tracking control**: “enable gaze control”, “calibrate gaze”, then click where your eyes rest.

---

## 13. The ELI server & using it from your phone

[Some setup] *Slightly technical, but only at setup.*

ELI includes a built-in **server** and **web app**. Start the server on your computer and chat with ELI from your **phone or tablet’s browser** over your home Wi-Fi — while all the actual thinking stays on your computer.

flowchart TD

```

A([Start the ELI Server<br/>on your computer]) --> B{Who should reach it?}
B -->|Just this computer<br/>default| C[Private to localhost]
B -->|My phone/tablet<br/>on home Wi-Fi| D[Start in LAN mode]
D --> E[It prints an address<br/>+ a security token]
E --> F[Open that address in<br/>your phone's browser]
F --> G[Enter the token once]
G --> H([Chat with ELI from your phone<br/>thinking stays on the PC])
C --> H

```

- Start it from the desktop app — **Settings → Web Server** (a “this computer” button and a “phone / Wi-Fi” button that shows the exact link to open). Or run `scripts/eli_serve.sh` (Mac/Linux) / `scripts/eli_serve.ps1` (Windows). Full details: `docs/SERVER_AND_WEB_APP.md`.
- The web app is **installable** — your phone’s “Add to Home Screen” makes it open like an app, and the shell works offline. Light/dark themes; you can recolour the chat.

### Starting the server, and what it prints

Start the server and it does two friendly things so you never have to hunt for the address: it **prints a clickable link** (Ctrl/Cmd-click it right in the terminal), and it **auto-opens your local browser** to the

dashboard. (Set `ELI_API_NO_BROWSER=1` if you'd rather it didn't.)

- **This computer only (default):** the dashboard opens at `http://127.0.0.1:8081/`. No token — you're on the machine, so you're trusted automatically.
- **Phone / Wi-Fi (LAN mode):** it prints your computer's LAN address (e.g. `http://192.168.1.20:8081/`) with a **one-time token** baked into the link, plus a **QR code** in the Connect tab. Scan it or open the link and you're in.

### The token, and the rotate button

The phone's token is **stable** — it's saved (permissions `0600`, under your config dir) and survives server restarts, so a paired phone stays paired; no re-scanning every reboot. Lost the phone, or just want a clean slate? Hit **rotate**: it mints a fresh token and instantly cuts off anything still using the old one.

### How the link keeps your token safe

The token rides in the link's **fragment** — the part after the `#` (`...:8081/#token=...`). Browsers **never send the fragment to the server**, so your token can't land in the server's access log (or any proxy log in between). When the page loads it reads the token from the fragment, stores it locally, then **wipes it from the address bar** so it isn't left in your history either. Older links that used `?token=` still work — the page accepts both — but everything ELI prints or displays now uses the safe fragment form, on the desktop panel and both launcher scripts alike.

### Voice from the phone needs HTTPS

Browsers block the microphone on a plain `http://` LAN address, so the phone **mic** needs a secure page. Start with `--https` (or the HTTPS toggle) and ELI runs a *second* server alongside on port **8443** with a self-signed certificate — accept the one-time “not private” warning and the mic works. Plain HTTP stays the primary path (it's what QR scanners open reliably); HTTPS runs *in addition*, purely for voice.

### Switching the model from the dashboard

**Settings → Model** lists every model installed on your computer, with sizes. Pick one and ELI **hot-swaps** it live — no restart. Only real, installed files show up, so you can't point it at something that won't load, and only an **admin** can switch it.

### Two servers, one machine

Worth knowing: “the server” is really **two** local services. The **web/API server** (port 8081) serves the dashboard, the phone app, and a small API. The **device server** is a separate MQTT service that talks to your smart-home gear (see the Home tab, below). Both run locally; both are yours. Everything about who's allowed in — tokens, roles, the loopback trust — is laid out in **Appendix C (Security)**.

### Under the hood (for scripts & power users)

The web/API server is a **FastAPI** app served by **uvicorn** (`api/server.py`). You rarely touch it directly — the desktop app and the launcher scripts wire everything up — but if you script it, a handful of environment variables control it: `- ELI_API_HOST` — `127.0.0.1` (default, this-computer-only) or `0.0.0.0` (LAN mode). `- ELI_API_PORT` — the HTTP port (default 8081); the HTTPS voice server uses 8443. `- ELI_API_TOKEN` — set your own token; if unset, ELI uses the saved stable one (`config/api_token`, permissions `0600`) so a paired phone survives restarts. `- ELI_API_ALLOW_TOKENLESS` — honoured only on loopback; LAN mode always requires the token. `- ELI_API_NO_BROWSER=1` — start without auto-opening the local browser.

The launchers wrap all of this: `scripts/eli_serve.sh` (Mac/Linux) and `scripts/eli_serve.ps1` (Windows) start LAN mode, then print the phone link (with the safe `#token=` fragment) and the **exact** firewall command for your OS — the quickest way in from a terminal. The API itself is a small REST surface

(chat, status/telemetry, model list/switch, voice, memory, research, audit); full endpoint detail lives in docs/SERVER\_AND\_WEB\_APP.md.

### If a phone can't connect

Nine times out of ten it's the firewall. When it starts in LAN mode ELI prints the **exact** command to open the port for your OS — run that, and make sure the phone is on the **same Wi-Fi**. It's all scoped to your local network; nothing is exposed to the internet.

### What's in the web app

It has a sidebar with several sections: - **Overview** — a live dashboard (clock, GPU/CPU/RAM gauges, status, recent activity). - **Chat** — streamed replies with markdown/code, saved sessions, stop/regenerate, and voice in/out. - **Commands** — a searchable list of everything ELI can do. - **Home** — ELI's own **home-AI** (see below). - **System** — live, measured GPU/CPU/RAM/model telemetry. - **Research** — shared document corpora you can build and query **together**, with citations. - **Audit** — a tamper-evident log of every action. - **Admin** — (admin only) per-user activity, the risk-gate policy, and user management.

### Roles (optional)

By default the person on the computer is the operator (full admin). If others share it, you can create accounts with roles — **viewer** (read-only dashboards), **member** (use everything), **admin** (also the Admin console) — in the Admin tab. Each gets a one-time link; the audit trail then attributes actions to the right person.

### The Home tab — ELI as your home assistant

ELI controls smart devices **directly over MQTT** (ESPHome / Tasmota / Zigbee2MQTT) — no Home Assistant, no cloud: - **Find devices** on your network with one click (mDNS), or add one by its topics. - **Control** lights/switches, group them into **rooms**, and run room-wide on/off — from the app or by voice (“turn on the desk lamp”, “turn off the kitchen”). - **Scenes** — save a set of device states as “Movie mode” and activate it (by voice too). - **Automations** — *when* something happens (a time, **sunrise/sunset**, or a device turning on/off), *do* something (control a device or run a scene), optionally *only if* a condition holds. - ELI **learns** how you use your home and **suggests** automations you can accept with one tap.

---

## 14. Models: picking, switching, and training your own

[Some setup] The **model** is the “brain” ELI thinks with — a file on your computer. Bigger = smarter but needs more graphics memory (VRAM). ELI picks a suitable one at install; change it anytime.

### Which model should I get?

flowchart TD

```
A([How much graphics memory<br/>VRAM do you have?]) --> B{Amount?}
B -->|None / CPU only| C[qwen2.5-3b<br/>small and fast]
B -->|~8 GB| D[qwen2.5-7b default<br/>or qwen3-8b]
B -->|~12 GB| E[falcon3-10b<br/>or phi-4]
B -->|24 GB+| F[qwen3.6-35b-a3b<br/>or falcon-h1-34b]
C --> G([ELI auto-sizes it to your<br/>hardware — you tune nothing])
D --> G
E --> G
F --> G
```

## Downloading

- **Pick from a menu (choose any number of models):** `python -m eli.core.model_download --choose` — this is what the installer uses; you tick the ones you want (or all / auto / none).
- Let ELI choose one best-fit for your hardware: `python -m eli.core.model_download --auto`
- See the menu without downloading: `python -m eli.core.model_download --list`
- Grab a specific one: `python -m eli.core.model_download qwen2.5-7b`

You're not limited to one — download several and switch between them anytime (\$"Switching models").

| Command name                  | Model                              | Roughly needs          |
|-------------------------------|------------------------------------|------------------------|
| qwen2.5-3b                    | small & fast                       | 4 GB graphics / or CPU |
| qwen2.5-7b ( <i>default</i> ) | great all-rounder                  | 8 GB graphics          |
| qwen3-8b                      | newer, big context, good reasoning | 8 GB graphics          |
| falcon3-10b                   | a step up                          | 12 GB graphics         |
| phi-4                         | strong reasoning for its size      | 12 GB graphics         |
| qwen3.6-35b-a3b               | very capable (mixture-of-experts)  | 24 GB / or slow on CPU |
| falcon-h1-34b                 | largest option                     | 24 GB / or slow on CPU |

You can also just **drop any .gguf file** into the `models/` folder and ELI finds it.

## Switching models

In **Settings → Model** (or "Model menu / Auto Detect"), pick the file. ELI sizes it to your hardware automatically — you don't tune anything.

## Using Ollama instead (Windows, macOS, Linux)

[Some setup] If you already run **Ollama**, ELI can use your Ollama models instead of a GGUF file — handy when you've curated models there, or want ELI to share one Ollama install.

1. Install and start Ollama (<https://ollama.com/download>). On Windows and macOS it starts itself; on Linux use `systemctl --user start ollama` (or `ollama serve`).
2. Pull at least one model, e.g. `ollama pull llama3.2`.
3. In ELI: **Settings → Model → Backend = Ollama**, then **Refresh Ollama Models** and pick one. It's also in the startup model picker and the toolbar dropdown.

**Host and port.** Leave *Host* blank/default for a normal local install. Set it if Ollama runs elsewhere — a different port (`localhost:9999`), another machine (`192.168.1.20`), or a container. You can type it any way that makes sense (`localhost:11434`, `127.0.0.1:11434`, a bare IP) — ELI fills in the missing pieces. **Ollama on another machine is supported:** ELI treats a host you configured as a deliberate local service, so its offline-by-default guard won't block it and the Net toggle can stay off.

**If ELI can't see your models:** confirm Ollama is running (`ollama list` in a terminal). If it's on another machine, start it there with `OLLAMA_HOST=0.0.0.0:11434` so it accepts connections beyond that machine, and allow port 11434 through its firewall.

## [Advanced] Training your own ELI-flavoured model (advanced, optional)

You can fine-tune a model so it speaks in **ELI's voice** out of the box.

**See `blueprints/finetuning_guide.md`** — a full step-by-step from choosing a base model, through extracting a voice dataset and training, to producing a ready-to-load .gguf. Crucially, ELI's **personality stays live** (never frozen into the model); training only shapes the *manner* of speech.

## 15. How hard ELI thinks (reasoning modes)

[Everyone] ELI has five effort levels. Faster modes answer quickly; deeper modes think longer and use more of ELI’s internal helpers for hard problems.

| Mode            | Feel               | Use it for                              |
|-----------------|--------------------|-----------------------------------------|
| <b>quick</b>    | snappy             | greetings, simple commands, quick facts |
| <b>normal</b>   | balanced (default) | everyday questions and tasks            |
| <b>advanced</b> | more thorough      | tricky, multi-step asks                 |
| <b>research</b> | digs deep          | gathering and synthesising a lot        |
| <b>expert</b>   | maximum effort     | the hardest problems                    |

Ask “what reasoning mode are you in?” or “explain all your reasoning modes”. For most people the default is right — ELI also deepens on its own when a problem clearly needs it.

## 16. Settings you’ll actually touch

[Some setup] Most people never need these. The few that matter:

- **Model** — which brain ELI uses (§14).
- **Voice** — wake word, text-to-speech on/off, microphone.
- **Hardware / startup** — ELI auto-tunes; advanced users can nudge how much graphics memory to hold back, or how much context to use. Leave alone unless you know why.
- **Privacy** — ELI is offline by default; this is where anything network-related lives.

Two graphics cards? ELI can use both — set in the GPU profile settings; single-card just works.

## 17. Privacy & staying offline

[Everyone]

- **Nothing leaves your computer** unless you ask for something online (web search, news, weather, downloading a model). ELI tells you when it goes online.
- **Your data is yours** — conversations, memories, files, profile live in local databases. Delete anytime.
- **No accounts, no telemetry, no subscription.**
- A fresh install is a **blank slate** — ELI starts knowing nothing about you.

## 18. Troubleshooting

[Some setup] Most issues map to one of these. When in doubt, **ask ELI** — “run a full runtime audit” or “status” — it reports honestly.

### “ELI is replying very slowly”

flowchart TD

```
A([Replies feel slow]) --> B[Ask: 'what are you running on?']
B --> C{Running on GPU<br/>or CPU?}
C -->|CPU| D[Reinstall with graphics support,<br/>or pick a smaller model]
C -->|GPU but big model| E[Switch to a smaller model<br/>see section 14]
C -->|GPU, right-sized| F[That's normal speed for<br/>your hardware]
D --> G([Faster])
E --> G
F --> G
```

## Everything else

| Problem                                | Try this                                                                                                        |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Replies look like gibberish</b>     | A model whose context is too small. Pick a recommended one (§14); ELI sizes context automatically.              |
| <b>“I couldn’t find a model”</b>       | <code>python -m eli.core.model_download --auto</code>                                                           |
| <b>Memory/recall seems empty</b>       | Normal on a fresh install. Ensure the embedder downloaded: <code>python -m eli.core.model_download --aux</code> |
| <b>Voice/wake word not responding</b>  | “run voice diagnostics”; check the microphone in Settings → Voice.                                              |
| <b>Won’t go online for news/search</b> | By design (offline-first). The network is allowed per-request; check Settings → Privacy.                        |
| <b>A habit didn’t run</b>              | Open <b>Habits</b> — confirm it’s enabled and the time is right; or “show my habits”.                           |
| <b>Something seems broken</b>          | “run a full runtime audit” or “status”.                                                                         |

## 19. Quick phrasebook (cheat sheet)

**Everyday** - “open spotify and play lo-fi” · “pause” · “skip” - “what’s on my screen?” · “summarise report.pdf” - “set an alarm for 7am” · “set a timer for 10 minutes” - “what’s the news?” · “weather in Dublin” · “what time is it?”

**Memory & you** - “remember that ...” · “what do you know about me?” · “my name is Alex”

**Habits & proactive** - “show my habits” · “yes” / “no” (to an offer) · “morning report” · “start proactive mode”

**Writing & code** - “write a report on X” · “write a script to ...” · “fix the bugs in foo.py”

**Voice** - wake word “computer” + request · “change the wake word to athena” · “start dictation” · “say good morning” · “train my voice”

**Check on ELI** - “what are you running on?” · “status” · “what have you been working on?”

**Models (in a terminal)** - `python -m eli.core.model_download --list` · `python -m eli.core.model_download --auto`

## 20. Glossary

| Term                | Plain meaning                                                                       |
|---------------------|-------------------------------------------------------------------------------------|
| <b>Model / GGUF</b> | The “brain” file ELI thinks with. .gguf is just its file format.                    |
| <b>VRAM</b>         | Memory on your graphics card. More VRAM → you can run a bigger, smarter model.      |
| <b>Embedder</b>     | A tiny helper model that powers ELI’s memory/recall. Installed automatically.       |
| <b>Context</b>      | How much of the conversation ELI can “hold in mind” at once. ELI sets this for you. |
| <b>Vision model</b> | The model that lets ELI understand screens and images.                              |



| Term                      | Plain meaning                                                           |
|---------------------------|-------------------------------------------------------------------------|
| <b>Wake word</b>          | The phrase that makes ELI start listening (default “computer”).         |
| <b>Proactive mode</b>     | ELI doing helpful things on its own (briefings, suggestions).           |
| <b>Habit</b>              | A routine ELI learned and you approved, run on a schedule.              |
| <b>Fine-tuning</b>        | Optional advanced step to teach a model ELI’s voice (§14).              |
| <b>Offline-by-default</b> | ELI never uses the internet unless you ask for something that needs it. |

## 21. Appendix A — Under the hood: how ELI thinks (the full pipeline)

[Advanced] *For the curious / technical. You never need this to use ELI — but here is the complete, verified path from your words to ELI’s reply.*

Every request runs through one funnel — `CognitiveEngine.process()` — which takes one of three paths depending on what you asked and the reasoning mode. The diagram is the **real** pipeline (verified against the running code): a deterministic fast-path for plain commands, a **quick** path via the Agent Bus (the default), and a **deep** 12-stage Orchestrator for hard work.

flowchart TD

```

    U([Your input – text or voice]) --> R[Router<br/>regex priority pipeline<br/>+ LLM-intent fallback]
    R --> P[CognitiveEngine.process<br/>action · args · confidence · mode]
    P --> FP{Plain deterministic command?<br/>open app · media · volume · status · job}
    FP -->|Yes · PHASE45 fast-path| FX[execute_action] --> VERB([Reply returned verbatim<br/>no LLM, instant])
    FP -->|No| MODE{Reasoning mode}

    MODE -->|quick · default| BUS[Agent Bus dispatch]
    BUS --> AG[15 specialist agents run over a DAG<br/>memory · knowledge-graph · system · file-code<br/>intent]
    AG --> DR[DispatchResult<br/>grounding_confidence · agents_used · memory_context]
    DR --> GE{Low grounding on a<br/>factual question?}
    GE -->|Yes| ESC[Escalate: web tier / local tier / honest hedge] --> GOV
    GE -->|No| KIND{What kind of result?}
    KIND -->|self-contained action| GOV
    KIND -->|grounded control| GSYN[Grounded synthesis] --> GOV
    KIND -->|conversation / CHAT| SYS[Build enhanced system prompt<br/>persona + memory brief + context] --> GOV

    MODE -->|normal/advanced/research/expert · deep| ORCH[12-stage Orchestrator]

    subgraph DEEP[The 12-stage Orchestrator]
        direction TB
        01[1 · Intent routing] --> 02[2 · Persona lock]
        02 --> 03[3 · HyDE query expansion]
        03 --> 04[4 · Planner – channels + budgets for the mode]
        04 --> 05[5-7 · Parallel retrieval<br/>keyword · FTS5 turns · FAISS vector · RAG · KG]
        05 --> 08[8 · Hybrid merge]
        08 --> 09[9 · Cross-encoder rerank]
        09 --> 010[10 · Context assembly]
        010 --> 0105[10.5 · Persona handoff]
        0105 --> 011[11 · LLM generation]
    end
    end

```

```

ORCH --> 01
011 --> 012{12 · Confidence vs threshold}
012 -->|pass| GOV
012 -->|too low| REPAIR[Repair / regenerate] --> GOV

GEN[LLM generation<br/>broker.infer / stream] --> GOV[Output governor<br/>+ No-Fake-Actions guard<br/>n
GOV --> OUT([Reply to you])
VERB --> OUT

```

**The retrieval stages (5-7) are where memory plugs in** — that’s Appendix B. The two guarantees baked into the end of every path: the **Output governor** cleans the reply, and the **No-Fake-Actions guard** ensures ELI never claims it did something it didn’t.

### A.1 — The router (before any path is chosen)

Your words first hit the **router**. It runs a **regex-first priority pipeline**: an ordered set of fast, exact checks (web realtime lookups, media controls, file/PDF, memory, OS control, ...), each carefully shaped so it only fires on a real command — e.g. media controls need a whole word in command shape, so “metabolise” never triggers a media “tab”. If nothing matches, an **LLM intent classifier** is the fallback. The router emits {action, args, confidence} and hands off to `CognitiveEngine.process()`. The router also remembers the *last used path* so follow-ups (“do it again”, “the second one”) resolve correctly.

### A.2 — The three paths, and when each is taken

- **Fast-path (PHASE45) — no AI involved.** Plain, deterministic actions (open an app, pause music, set volume, report status, check a job) are executed directly and the result is returned **verbatim**. There’s no model call, so these are instant and can’t be “hallucinated”. This is why “pause” never produces a chatty paragraph.
- **Quick path (the default) — the Agent Bus.** Conversational and lightly-grounded requests go to the **Agent Bus**, which runs up to **15 specialist agents** in parallel over a dependency graph, each contributing evidence (memory hits, knowledge-graph facts, system readings, code search, capability info, ...). Their combined result carries a **grounding confidence** — an honest “is this actually backed by anything?” score.
- **Deep path — the 12-stage Orchestrator.** When you ask for depth (advanced/research/expert modes, or a hard question), ELI runs the full pipeline below: more retrieval, reranking, and a confidence check with automatic repair.

### A.3 — The Agent Bus, in full

The bus **selects a relevant subset** of agents for your intent (or fans out broadly when unsure), runs them concurrently, and **aggregates** their findings into a `DispatchResult`. Key agents and what they bring:

| Agent                                                | Contributes                                                        |
|------------------------------------------------------|--------------------------------------------------------------------|
| <b>Memory</b>                                        | recalls facts/turns from SQLite + full-text + vector search        |
| <b>Knowledge-Graph</b>                               | entities and how they relate (who/what/where)                      |
| <b>System</b>                                        | runs executor actions (web search, audits, status)                 |
| <b>File-Code</b>                                     | searches the actual codebase for grounded answers about ELI itself |
| <b>Introspection</b>                                 | live runtime/self state                                            |
| <b>Capability</b>                                    | what ELI can do (the capability manifest)                          |
| <b>Habit / Proactive / Reflection / Self-Improve</b> | routines, signals, insights, fixes                                 |
| <b>Plugin / Voice / Frontier / Orchestrator</b>      | plugin dispatch, speech state, deeper reasoning hooks, planning    |

After the bus returns, a **grounding-escalation hook** checks: *is this a factual question that came back weakly supported?* If so, ELI escalates — a web tier (if you allow online), a local tier, or an **honest hedge** (“I’m not certain, but...”) rather than a confident guess. Then the result is finalised: a self-contained action returns directly, a grounded control action gets a short grounded synthesis, and a conversation goes to generation with the full persona + memory context.

#### A.4 — The 12 stages, one by one (the deep path)

1. **Intent routing** — settle exactly what you’re asking for.
2. **Persona lock** — fix ELI’s identity/voice for this turn so it stays consistent.
3. **HyDE query expansion** — ELI drafts a *hypothetical answer* and searches with **that** too, which dramatically improves what memory retrieves (you find better matches by searching with an answer-shaped query, not just the bare question). Skipped for very short queries.
4. **Planner** — decide *which* retrieval channels to use and how big the budgets are, tuned to the reasoning mode (fast / balanced / deep). 5–7. **Parallel retrieval** — several searches run at once: keyword, **full-text (FTS5)** over your conversation history, **FAISS vector** (semantic) search, optional RAG, and the **knowledge graph**. (This is the seam where Appendix B’s memory system feeds in.)
5. **Hybrid merge** — combine the hits from every channel into one candidate set.
6. **Cross-encoder rerank** — a smarter model re-scores the merged candidates for true relevance to your question, and the best rise to the top. (Skipped for non-conversational asks.)
7. **Context assembly** — build the grounded context block that will inform the answer. 10.5. **Persona handoff** — wrap that context with ELI’s live persona brief for generation.
8. **LLM generation** — the model writes the reply (streaming or one-shot). For the deepest modes this is a two-part private-reasoning-then-condense step.
9. **Confidence check** — score the answer against a threshold; if it’s too low, ELI **repairs/regenerates** rather than shipping a weak answer.

#### A.5 — The two end-of-path guarantees

Every path, no matter which, ends at: - **Output governor** — normalises and cleans the reply (strips artefacts, enforces formatting). - **No-Fake-Actions guard** — if ELI’s reply *claims* it did something (played a song, fixed a file), the system verifies it actually happened; if not, the claim is replaced with the honest truth. This is enforced in code, not left to the model’s goodwill.

## 22. Appendix B — Under the hood: how ELI remembers (the full memory system)

[Advanced] *For the curious / technical. ELI’s memory is a genuine hybrid — relational + full-text + vector + graph — all local. This is the complete input → storage → recall path (verified against the code).*

flowchart TD

```

subgraph IN[INPUT – everything ELI may write]
    direction LR
    I1[Conversation turns]
    I2["'Remember X' durable facts"]
    I3[Observations + user patterns]
    I4[Habits + habit events]
    I5[Failures · corrections · improvements]
    I6[Knowledge-graph entities + relations]
    I7[User Model brief]
end
IN --> MEM[Memory class<br/>memory.py routes every write]

MEM --> D1[(user.sqlite3<br/>memories · turns · habits<br/>patterns · user_model · KG)]

```

```

MEM --> D2[(agent.sqlite3<br/>dispatch + metrics)]
MEM --> D3[(system_index.sqlite3<br/>apps · files · dirs)]
MEM --> D4[(coding_memory.sqlite3<br/>bug fixes)]
MEM -->|embeds new text| V[[FAISS vector index<br/>nonomic embedder<br/>serialized – segfault-safe]]
MEM -->|indexes text| F[[FTS5 full-text<br/>memories + turns + KG]]
MEM -->|entities/relations| KG[[Knowledge graph<br/>subject - predicate - object]]

Q([Recall request<br/>from pipeline stages 5-7]) --> RC[recall_memory<br/>hybrid retriever]
V --> RC
F --> RC
RC --> S1[1 · FAISS vector search – primary]
S1 --> S2{Cold index or<br/>fewer than limit/2 hits?}
S2 -->|Yes| S3[2 · FTS5 / LIKE keyword supplement]
S2 -->|No| S4
S3 --> S4[3 · Noise filter<br/>drop ELI's own insights / reflections /<br/>session summaries / rows over]
S4 --> S5[4 · Importance-weighted ordering<br/>COALESCE importance 0.5]
S5 --> CTX([Grounded context handed to the prompt])
KG -->|context_for_prompt| CTX
I7 -->|read fresh every turn| CTX

subgraph MECH[UPKEEP MECHANISMS]
  direction LR
  M1[Weight decay<br/>old, low-importance rows fade ×0.98]
  M2[Embedder lock<br/>one-at-a-time embedding]
  M3[recall_log<br/>records what was recalled]
end
M1 --> D1

```

**Why it’s trustworthy:** the **noise filter** stops ELI’s own past replies and reflections from resurfacing as if they were *your* facts; the **embedder lock** keeps the vector engine from crashing under concurrent use; and **weight decay** is the gentle “forgetting” that lets stale topics fade while active ones stay fresh. Nothing here leaves your computer.

## B.1 — What ELI writes (the inputs)

Everything that flows into memory passes through one **Memory** module, which decides where each item belongs: - **Conversation turns** — what you said and what ELI replied, with timestamps. - **Durable facts** — anything you explicitly ask it to “remember”. - **Observations & user patterns** — quiet signals about your interests, focus, and tone that build the living profile. - **Habits & habit events** — your repeated actions and the routines learned from them. - **Failures, corrections, improvements** — ELI’s own mistakes and fixes, kept for self-learning (these are deliberately **excluded** from normal recall — see the noise filter). - **Knowledge-graph entities & relations** — named things and how they connect. - **User Model brief** — a compact, pre-rendered summary of who you are, refreshed over time.

## B.2 — Where it’s stored (four local databases + three indexes)

Everything is plain local files on your machine — no server, no cloud: - **user.sqlite3** — the main store: memories, conversation turns, habits, patterns, the User Model, and the knowledge graph. - **agent.sqlite3** — ELI’s internal agent activity and performance metrics. - **system\_index.sqlite3** — an index of your apps, executables, files, and folders (so “open the file manager” is instant). - **coding\_memory.sqlite3** — remembered code bug-fixes.

On top of the relational data sit three search indexes: - **FAISS vector index** — every stored text is turned into an “embedding” (a list of numbers capturing meaning) by a small local **nonomic embedder**, so ELI can find things by *meaning*, not just exact words. Embedding is done **one-at-a-time** behind a lock (the embedder isn’t thread-safe; serialising it prevents crashes). - **FTS5 full-text index** — fast exact/keyword

search over memories, conversation turns, and the knowledge graph. - **Knowledge graph** — a subject → predicate → object web (e.g. *you* → *own* → *a Tesla*) that ELI can walk to answer relational questions.

### B.3 — How recall works (step by step)

When the cognition pipeline (stages 5–7) needs memory, it calls **recall\_memory**, a genuine hybrid retriever that runs in this order: 1. **Vector search first (FAISS)**. The semantic index is the primary path — it finds things that *mean* the same even if worded differently. 2. **Keyword supplement (FTS5/LIKE) — only if needed**. If the vector index is cold/empty or returns fewer than half the requested results (e.g. a very short query), a keyword search tops it up. (When another part of the system already did the vector search, this step is skipped to avoid double-searching.) 3. **Noise filter**. Results are scrubbed of ELI’s *own* output — assistant insights, reflections, session summaries, the “orchestrator” source, and any over-long blobs (>1500 chars). This is the safeguard that stops ELI’s past musings from masquerading as *your* facts. 4. **Importance-weighted ordering**. What survives is ranked by an importance score so the most significant memories surface first.

Alongside this, the **knowledge graph** contributes a lightweight relational context, and your **User Model brief** is read **fresh every single turn** — that’s why ELI stays in context about who you are without you repeating yourself.

### B.4 — Upkeep (the mechanisms that keep memory healthy)

- **Weight decay** — old, low-importance memories are gently faded (multiplied down over time), so abandoned topics naturally recede while active ones stay prominent. This is ELI’s “forgetting”.
- **Embedder serialization** — the single lock around the embedder that keeps the vector engine stable under concurrent access.
- **Recall log** — a record of what was retrieved, used for tuning and for honestly explaining *where* a piece of knowledge came from (“how do you know that?”).

### B.5 — Privacy, restated

Every database and index in this appendix is a local file under your user profile. There is no sync, no upload, no account. Delete the files (or use the **Memory** tab) and that knowledge is gone. A fresh install starts with the full structure but **zero** personal rows — a true blank slate.

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## 23. Appendix C — Under the hood: how ELI stays safe and yours (security)

[Advanced] *For the curious / technical. ELI is meant to be safe by construction, not by good intentions — every guard below is enforced in code, not left to a setting you might forget to flip. All of this is verified against the running codebase.*

### C.1 — Offline by default (the socket failsafe)

ELI doesn’t just *prefer* to stay offline — it’s physically stopped from reaching the internet unless you asked for something that needs it. At startup, netguard installs a **process-wide socket guard**: it patches the low-level `connect()` call so that *any* outbound connection to a non-loopback address raises an `OfflineError` while ELI is offline. Loopback and local sockets (the phone web server, the local model) always work. So even if some stray bit of code *tried* to phone home, the operating-system call itself is blocked — belt and braces. When you do turn networking on for a task, that flip is written to the audit ledger; “internet on” is never silent.

Ask for something online while offline and you don’t get a crash or an invented answer — you get an honest refusal: “*I can’t fetch that right now, I’m offline.*”

## C.2 — The fail-closed command gate

ELI can run shell commands and open apps, so that path is gated hard. Before any command runs it's checked against an **allowlist** by the SecurityManager. The part that matters: if the security layer can't load for any reason, the check **fails closed** — it returns *denied*, not *allowed*. A broken guard blocks the action; it never quietly waves one through. Sandboxed runs are limited to a fixed set of allowed executables.

## C.3 — The approval gate (nothing risky happens behind your back)

Anything ELI proposes on its own — an autonomous action, a self-improvement patch, a scheduled job — goes through the **approval engine** before it can execute. Each proposal carries an *action class*, and the engine decides: auto-approve the harmless ones, or hold the riskier ones until **you** confirm. Who is even allowed to propose what is also gated — a background daemon can't quietly emit a dangerous action. There's a **Full Control** mode that lifts the confirmation barrier if you want ELI to just act, but that's your explicit choice, and it's off by default.

## C.4 — The phone: who's allowed in

The web server (§13) is locked down. On the same machine (loopback) you're trusted automatically. From the network, every request must carry a **bearer token** — and that token is never stored in plain text, only its SHA-256 hash. The token travels in the link's URL **fragment** (after the #), which browsers never transmit to the server, so it can't appear in the access log; the page reads it locally and immediately clears it from the address bar. Define one or more users and **RBAC** switches on: each token maps to a role (admin / member), and admin-only actions — changing settings, switching the model — reject anyone who isn't admin. With no users defined it's single-operator mode: just you. The **rotate** button issues a fresh token and kills the old one, so a paired phone can be cut off in one tap.

## C.5 — The audit ledger

Security-relevant events — settings changes, the network being turned on or off, research collaborations — are written to a **tamper-evident audit ledger**: a local, append-only trail so there's always an honest record of what changed and when.

## C.6 — Privacy, restated

Every guard, token, ledger, and database above is a local file under your own user profile. No account, no sync, no telemetry. ELI's default posture is *closed* — offline, gated, and yours. You open doors deliberately, one at a time.

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*That's the whole assistant — inside and out. You don't need any of the last three appendices to use ELI; just talk to it in plain English, and ask "what can you do?" whenever you're curious. Everything here happens on your computer, for you, privately.*